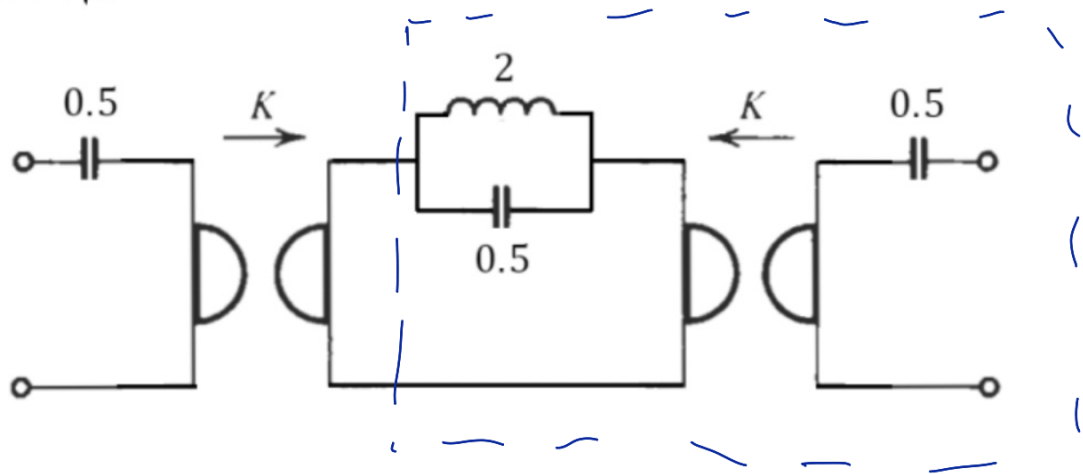


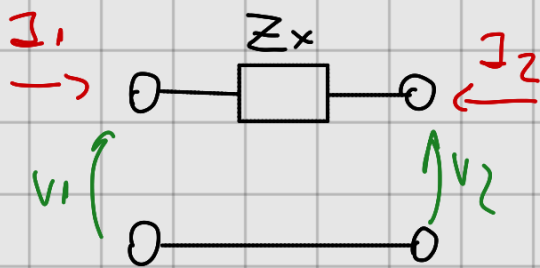
Ejercicio #6

Obtenga un circuito lattice equivalente al siguiente cuadripolo. Utilice la constante del girador $K = \sqrt{2}$



Trabajo con parámetros T.

comienzo con las siguientes definiciones,



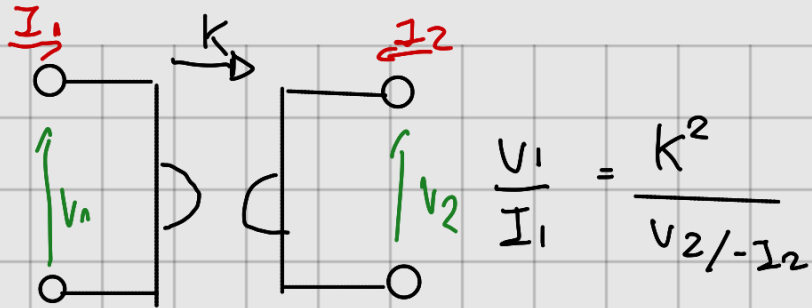
$$T_{xA} = \frac{V_1}{V_2} \Big|_{I_2=0} = 1$$

$$T_{xB} = \frac{V_1}{-I_2} \Big|_{V_2=0} = Z_x$$

$$T_{xC} = \frac{I_1}{V_2} \Big|_{I_2=0} = 0$$

$$T_x = \begin{pmatrix} 1 & Z_x \\ 0 & 1 \end{pmatrix}$$

$$T_{xD} = \frac{I_1}{-I_2} \Big|_{V_2=0} = 1$$



$$T_{G_A} = \left. \frac{V_1}{V_2} \right|_{I_2=0} = 0$$

$$T_{G_B} = \left. \frac{V_1}{-I_2} \right|_{V_2=0} = k$$

$$T_{G_C} = \left. \frac{I_1}{V_2} \right|_{I_2=0} = \frac{1}{k}$$

$$T_{G_D} = \left. \frac{I_1}{-I_2} \right|_{V_2=0} = 0$$

$$T_G = \begin{pmatrix} V_1 & V_2 & I_2 \\ I_1 & 0 & k \\ \frac{1}{k} & 0 & 0 \end{pmatrix}$$

con estas secciones, el T equivalente quedaría...

$$T_T = T_x \Big|_{Z_x = \frac{1}{sC}} \cdot T_G \cdot T_x \Big|_{Z_x = \frac{1}{sC} // sL} \cdot T_G \cdot T_x \Big|_{Z_x = \frac{1}{sC}}$$

$$T_T = \begin{pmatrix} 1 & Z_C \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & k \\ \frac{1}{k} & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & Z_C // Z_L \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & k \\ \frac{1}{k} & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & Z_C \\ 0 & 1 \end{pmatrix}$$

$$T_T = \begin{pmatrix} Z_C \cdot \frac{1}{k} & k \\ \frac{1}{k} & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & Z_C // Z_L \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & k \\ \frac{1}{k} & \frac{1}{k} \cdot Z_C \end{pmatrix}$$

$$T_T = \begin{pmatrix} Z_C \cdot \frac{1}{k} & \frac{Z_C}{k} \cdot (Z_C // Z_L) + k \\ \frac{1}{k} & \frac{1}{k} \cdot (Z_C // Z_L) \end{pmatrix} \begin{pmatrix} 1 & k \\ \frac{1}{k} & \frac{1}{k} \cdot Z_C \end{pmatrix}$$

$$T_T = \begin{pmatrix} \frac{Z_C}{k^2} \cdot (Z_C // Z_L) + 1 & 2 \cdot Z_C + \frac{Z_C^2}{k^2} (Z_C // Z_L) \\ \frac{1}{k^2} \cdot (Z_C // Z_L) & 1 + \frac{1}{k^2} \cdot Z_C \cdot (Z_C // Z_L) \end{pmatrix}$$

como $Z_C = \frac{2}{s}$, $Z_L = 2 \cdot s$, $k = \sqrt{2}$

y $Z_C // Z_L = \frac{L/C}{\frac{1}{sC} + sL} = \frac{sL}{s^2LC + 1} = \frac{s \cdot \frac{1}{C}}{s^2 + 1/C} = 2 \cdot \frac{s}{s^2 + 1}$

entonces...

$$T_T = \begin{pmatrix} \frac{1}{s} \cdot 2 \frac{\cancel{s}}{s^2+1} + 1 & \frac{4}{s} + \frac{2}{s^2} \cdot 2 \frac{\cancel{s}}{s^2+1} \\ \frac{1}{2} \cdot \cancel{2} \frac{s}{s^2+1} & 1 + \frac{1}{\cancel{s}} \cdot 2 \frac{\cancel{s}}{s^2+1} \end{pmatrix}$$

$$T_T = \begin{pmatrix} \frac{2}{s^2+1} + 1 & \frac{4}{s} + \frac{4}{s} \cdot \frac{1}{s^2+1} \\ \frac{s}{s^2+1} & 1 + \frac{2}{s^2+1} \end{pmatrix}$$